

RESULTS

Results

01 · WELCH'S ANOVA

3+ groups, unequal variances

Table 1. Descriptives by group

Group	N	Mean	Std. Dev.
Blended	80	65.49	6.85
Flipped	80	70.75	6.58
Lecture	80	60.46	7.57

Table 2. Welch's ANOVA

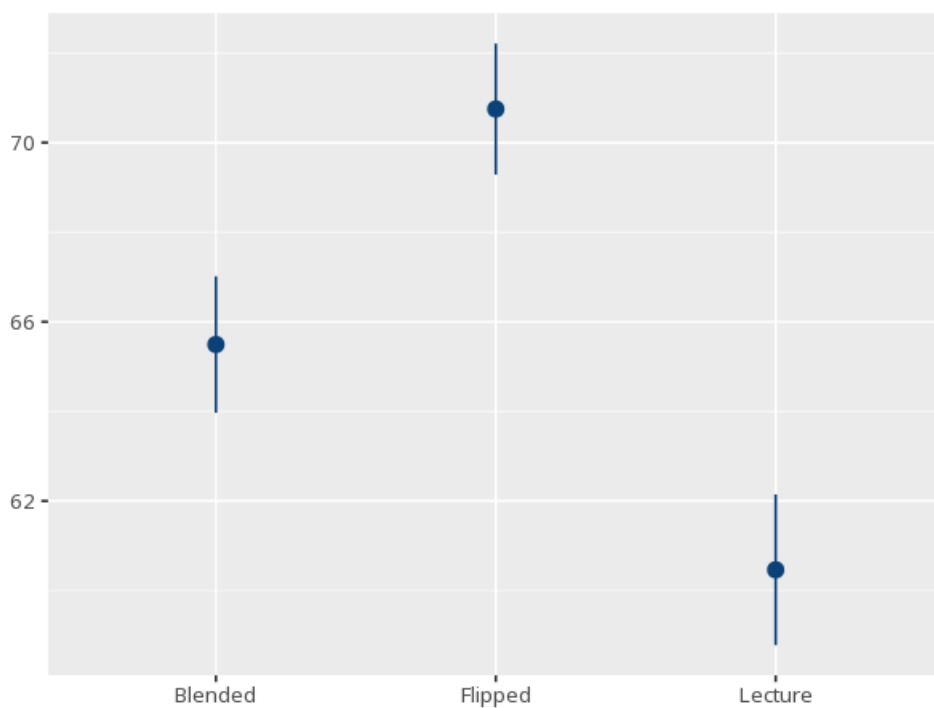
F	df1	df2	p	ω² [95% CI]
42.31	2	157.49	<.001	.34 [.24, 1.00]

Table 3. Games-Howell post-hoc

Pair	M _{diff}	p _{adj}	95% CI
Blended - Flipped	-5.26	<.001	[-7.77, -2.75]
Blended - Lecture	5.03	<.001	[2.33, 7.73]
Flipped - Lecture	10.29	<.001	[7.63, 12.94]

Welch's adjusts the degrees of freedom so equal variances are not assumed (df2 is fractional); within-group normality is still assumed and checked with Shapiro-Wilk per group. (Shapiro per group: Blended W=0.98, p=.271; Flipped W=0.98, p=.397; Lecture W=0.97, p=.088) - normality looks reasonable across groups

Figure. Group means



Understanding the numbers

F. The ratio of between-group to within-group variance, computed without assuming equal variances. Here, $F(2, 157.49) = 42.31$.

df1. The numerator degrees of freedom - one less than the number of groups. Here, $df1 = 2$.

df2. The (fractional) denominator degrees of freedom, adjusted downward when group variances are unequal. Here, $df2 = 157.49$.

p. The probability of an F this large if all group means were truly equal. Here, $p < .001$ - below the $\alpha = 0.05$ threshold.

ω^2 . The effect size for a Welch F-test - the estimated share of variance explained by group membership, converted directly from $F/df1/df2$ since Welch's test has no fitted model to compute it from. Here, $\omega^2 = .34$ [.24, 1.00].

Mean. The average outcome for that group. Here, each of the 3 groups has its own mean in the descriptives table above.

N. The number of observations in that group. Here, each of the 3 groups has its own N in the descriptives table above.

Std. Dev.. How spread out outcome scores are within that group. Here, each of the 3 groups has its own SD in the descriptives table above.

M diff. The difference between two group means in the Games-Howell post-hoc table. Here, each pairwise contrast above reports its own mean difference in the Games-Howell table.

p (adj). The pairwise p-value, Games-Howell-adjusted (variance-robust, like the omnibus test) for the number of comparisons. Here, each pairwise contrast above carries its own Games-Howell-adjusted p.

95% CI. The range likely to contain the true mean difference for that pair. Here, each contrast's 95% CI is shown in the Games-Howell table above.

How to read this test

A one-way ANOVA that relaxes the equal-variance assumption but still assumes roughly normal data within each group. A significant F means at least one group mean differs from the others; the Games-Howell post-hoc (also variance-robust) then shows which specific pairs differ. Your significance threshold (α) is 0.05.

APA template: Welch's ANOVA gave $F(2,157.49)=42.31$, $p < .001$, $\omega^2=.34$ [.24, 1.00].

Statistical basis: Welch's ANOVA (heteroscedasticity-robust F, fractional df) (Welch, B. L. (1951). "On the comparison of several mean values: an alternative approach." *Biometrika*, 38(3-4), 330-336.); Games-Howell post-hoc comparisons (also variance-robust) (Kassambara A (2023). *rstatix: Pipe-Friendly Framework for Basic Statistical Tests*. R package.); ω^2 effect size (F-to- ω^2 conversion, no fitted aov model needed) (Ben-Shachar MS, Lüdtke D, Makowski D (2020). "effectsize: Estimation of Effect Size Indices and Standardized Parameters." *Journal of Open Source Software*, 5(56), 2815.). Full references in the exported CITATIONS.txt.

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